# 1. Practice: phase portrait

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## Practice drawing phase portrait

4/4 points (graded)

The matrices that we will be studying will all have the form

$$\mathbf{A} = \begin{pmatrix} a & 2 \\ -2 & -1 \end{pmatrix} \quad \text{where } a \text{ is real.} \qquad (3.112)$$

Compute the trace, determinant, and characteristic polynomial in terms of $a$.

$\mathrm{tr}\mathbf{A} =$ a-1 ✓

$a-1$

Calculator Hide Notes

$\det\mathbf{A} =$ 4-a ✓

$4-a$

$\det(\mathbf{A}-\lambda\mathbf{I}) =$ lambda^2-(a-1)*lambda+(4-a) ✓

$\lambda^2-(a-1)\cdot\lambda+(4-a)$

For all such matrices $\mathbf{A}$, express the determinant as a function of the trace. (Enter **Tr** for $\mathrm{tr}\mathbf{A}$.)

$\det\mathbf{A} =$ 3-Tr ✓

$3-Tr$

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✓ Correct (4/4 points)

## The family of matrices on the trace determinant plane

1/1 point (graded)

As above, we are studying the family of matrices

$$\mathbf{A} = \begin{pmatrix} a & 2 \\ -2 & -1 \end{pmatrix}. \qquad (3.113)$$

On the trace-determinant plane below,

- sketch the critical parabola $\mathrm{Det} = (\mathrm{Tr})^2/4$,
- sketch the curve given by the relationship between the trace and determinant of all matrices of the given form,
- on this curve, plot the points corresponding to $a = -6, -5, -2, 1, 2, 3, 4, 5$.

(It may be easier to first plot the points. In particular, first find the intersection points of the critical parabola and the curve representing the given family of matrices.)

(Use the point tool labeled with an $a$ value to mark the corresponding point. You should have one point per point tool.
Use the spline tool labeled **curve of matrices** to sketch the curve given by the relationship between the trace and determinant of the matrices.
Use the spline tool labeled **critical parabola** ' to sketch the critical parabola.)

✓

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✓ Correct (1/1 point)

## Stability and phase portraits

8/8 points (graded)

As above, we are studying the family of matrices

$$\mathbf{A} = \begin{pmatrix} a & 2 \\ -2 & -1 \end{pmatrix}. \qquad (3.114)$$

For each of the following values of $a$, use the sketch in the previous problem to determine the stability, and type of the phase portrait of the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$.
determine the direction of any spiral or center.
(In each column below, check all that apply.)
(Note that due to space limitation, the choice with the single word **node** refers to **nodal sources/sinks**, not degenerate or star nodes.)

| $a=-6$ | $a=-5$ | $a=-2$ | $a=1$ |
|---|---|---|---|
| ☑ stable | ☑ stable | ☑ stable | ☐ stable |
| ☐ unstable | ☐ unstable | ☐ unstable | ☐ unstable |
| ☐ neutrally stable | ☐ neutrally stable | ☐ neutrally stable | ☑ neutrally stable |
| ☑ node | ☐ node | ☐ node | ☐ node |
| ☐ spiral | ☐ spiral | ☑ spiral | ☐ spiral |
| ☐ saddle | ☐ saddle | ☐ saddle | ☐ saddle |
| ☐ center | ☐ center | ☐ center | ☑ center |
| ☐ comb | ☐ comb | ☐ comb | ☐ comb |
| ☐ defective node | ☑ defective node | ☐ defective node | ☐ defective node |
| ☐ star node | ☐ star node | ☐ star node | ☐ star node |
| ☐ clockwise | ☐ clockwise | ☑ clockwise | ☑ clockwise |
| ☐ counter-clockwise | ☐ counter-clockwise | ☐ counter-clockwise | ☐ counter-clockwise |
| ✓ | ✓ | ✓ | ✓ |

| $a=2$ | $a=3$ | $a=4$ | $a=5$ |
|---|---|---|---|
| ☐ stable | ☐ stable | ☐ stable | ☐ stable |
| ☑ unstable | ☑ unstable | ☑ unstable | ☑ unstable |
| ☐ neutrally stable | ☐ neutrally stable | ☐ neutrally stable | ☐ neutrally stable |
| ☐ node | ☐ node | ☐ node | ☐ node |
| ☑ spiral | ☐ spiral | ☐ spiral | ☐ spiral |
| ☐ saddle | ☐ saddle | ☐ saddle | ☑ saddle |
| ☐ center | ☐ center | ☐ center | ☐ center |
| ☐ comb | ☐ comb | ☑ comb | ☐ comb |
| ☐ defective node | ☑ defective node | ☐ defective node | ☐ defective node |
| ☐ star node | ☐ star node | ☐ star node | ☐ star node |
| ☑ clockwise | ☐ clockwise | ☐ clockwise | ☐ clockwise |
| ☐ counter-clockwise | ☐ counter-clockwise | ☐ counter-clockwise | ☐ counter-clockwise |
| ✓ | ✓ | ✓ | ✓ |

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✓ Correct (8/8 points)

## Eigenvalues and more detailed phase portrait

2 points possible (graded)

Find the eigenvalues in terms of $a$ for the case when the eigenvalues are real and distinct.
(The order does not matter.)

$\lambda_1 =$

$\lambda_2 =$

Now, for more detailed phase portraits, on paper:

- If the eigenvalues are real, find a nonzero eigenline for each. In those cases, sketch the eigenlines and some trajectories not on eigenlines
- If the eigenvalues are not real, find the pseudoperiod of oscillation of the solution and the exponent in the term giving exponential growth or decay of the magnitude of the solution. Sketch the trajectories of a few real solutions.

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### How and why to use the forum

While answers to most individual problems are provided, staff do not provide detailed solutions to recitation problems. Instead, use the forum to collaborate with your fellow students to develop detailed solutions for these problems.

The goal is to get practice developing your own detailed solutions, practice explaining solutions to others, and read other people's solutions and reflect on their process and reasoning.

## 1. Practice: phase portrait

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### Defective Node

question posted 3 months ago by sandipvinayka

1. How to identify whether a node is defective or star?
2. Why isn't defective node associated with a direction (clockwise or counterclockwise)?

This post is visible to everyone.

Add a Response

1 response

**jfrench** (Staff)
3 months ago

Nodes aren't actually rotating! Instead there is a direction it is tangent from $-\infty$ and towards $+\infty$. But we don't consider it a rotation per se.

OK Thanks! Regarding defective node or star node it was told in recitation that if the matrix is diagonal it's a star else it's a defective node. However I didn't understand that conceptually. Any light on this would be welcome.

posted 3 months ago by sandipvinayka

Hi, this is how I understand it. These two systems lie both on the critical parabola in the $(tr, det)$ plane and both have one repeated eigenvalue $\lambda = 1$:

A) $\dot{\mathbf{x}} = \begin{pmatrix} 1 & amp;0 \\ 0 & amp;1 \end{pmatrix}\mathbf{x}$ (star node)

B) $\dot{\mathbf{x}} = \begin{pmatrix} 2 & amp;1 \\ -1 & amp;0 \end{pmatrix}\mathbf{x}$ (defective node)

A difference between the two is that A) has inifinitely many eigenlines and B) only one (you can verify this by determining the eigenvectors for both systems). In case of A), these many eigenlines form a star. This all means that in case of A), regardless of where your startpoint $(x_0 \neq 0, y_0 \neq 0)$ is, you will start on an eigenline and therefore move to infinity in a straight line, which starts in the critical point. You can also see from system A) that $\dot{x} = x$ and $\dot{y} = y$. So if you start for example on the positive y-axis, then $\dot{x}$ will always stay $0$, so you always move vertically up. If you start for example at $(1,1)$, then $\dot{x} = 1$ and $\dot{y} = 1$, so you will move to infinity on the line $y = x$.

In case of B) you move only in a straight line to infinity when $(x_0 \neq 0, y_0 \neq 0)$ lies on the eigenline, otherwise your trajectory will approach a vector parallel to the eigenline when going to infinity.

I hope this helps.

posted 3 months ago by MarkvdV

Excellent explanation @MarkvdV! :)

Geometrically, the reason the identify matrix (or multiple of the identify) is called a star exactly comes from the realization that every line is an eigenline. This means that the eigenlines look like rays emanating out of the origin. Hence "star".

posted 2 months ago by jfrench (Staff)

Thanks @MarkvdV. I think I understand this now.

posted 2 months ago by sandipvinayka

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